VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Practical No. 07

Aim: Write a Program to implement Simple and Multiple Linear Regression Models.

Software Required: Python IDE, Anaconda, VS Code, Jupyter Nootbook, Google Colab.

Theory:

Linear Regression

Linear regression is a type of supervised machine-learning algorithm that learns from the labelled datasets and maps the data points with most optimized linear functions which can be used for prediction on new datasets. It assumes that there is a linear relationship between the input and output, meaning the output changes at a constant rate as the input changes. This relationship is represented by a straight line.

For example, we want to predict a student's exam score based on how many hours they studied. We observe that as students study more hours, their scores go up. In the example of predicting exam scores based on hours studied. Here

- **Independent variable (input):** Hours studied because it's the factor we control or observe.
- **Dependent variable (output):** Exam score because it depends on how many hours were studied.

Types of Linear Regression

1) Simple Linear Regression

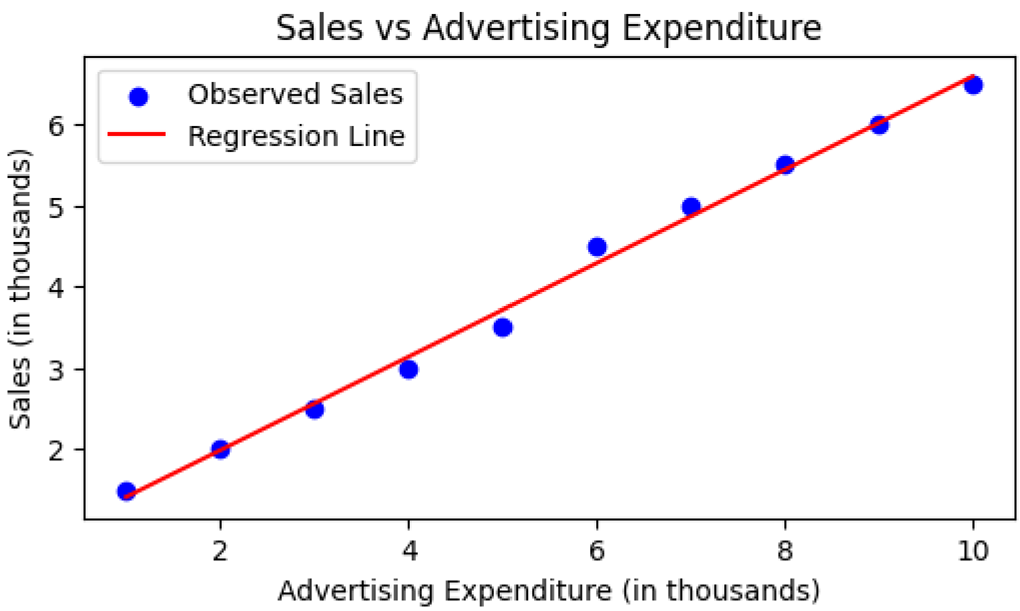
2) Multiple Linear Regression

1) Simple Linear Regression:

Simple linear regression models the relationship between a dependent variable and a single independent variable.

Simple Linear Regression aims to describe how one variable i.e. the dependent variable changes in relation with reference to the independent variable. For example, consider a scenario where a company wants to predict sales based on advertising expenditure. By using simple linear regression, the company can determine if an increase in advertising leads to higher sales or not.

The below graph explains the relationship between advertising expenditure and sales using simple linear regression:



Simple Linear Regression

The relationship between the dependent and independent variables is represented by the simple linear equation:

$y=mx+b$

- y is the predicted value (dependent variable).
- m is the slope of the line
- x is the independent variable.
- b is the y-intercept (the value of y when x is 0).

In this equation m signifies the slope of the line indicating how much y changes for a one-unit increase in x, a positive m suggests a direct relationship while a negative m indicates an inverse relationship.

To better understand this relationship, we can express it in a more statistical context using the linear regression formula:

$Y=\beta 0+\beta 1x$

In simple linear regression the parameters β0 and β1𝛽0 and 𝛽1 play crucial roles in defining the relationship between the independent variable X and the dependent variable Y in the regression equation.

2) Multiple Linear Regression

Multiple Linear Regression extends this concept by modelling the relationship between a dependent variable and two or more independent variables. This technique allows us to understand how multiple features collectively affect the outcomes.

Steps for Multiple Linear Regression

Steps to perform multiple linear regression are similar to that of simple linear Regression but difference comes in the evaluation process. We can use it to find out which factor has the highest influence on the predicted output and how different variables are related to each other. Equation for multiple linear regression is:

$y=\beta_0+\beta_1X_1+\beta_2X_2+\cdots+\beta_nX_n$

Where:

- y is the dependent variable
- $X_1,X_2,\cdots X_n$ are the independent variables
- β_0 is the intercept
- $\beta_1,\beta_2,\cdots\beta_n$ are the slopes

The goal of the algorithm is to find the best fit line equation that can predict the values based on the independent variables. A regression model learns from the dataset with known X and y values and uses it to predict y values for unknown X.

Advantages of Simple Linear Regression (SLR)

1. **Easy to interpret** – Only one predictor, so results are simple and intuitive.
2. **Quick to compute** – Requires less data and computation.
3. **Good for initial analysis** – Helps in identifying basic relationships between two variables.

Disadvantages of Simple Linear Regression (SLR)

1. **Limited scope** – Considers only one independent variable; ignores other influencing factors.
2. **Less accurate** – If multiple factors affect the outcome, predictions may be misleading.
3. **Assumptions sensitive** – Strong assumptions (linearity, independence, homoscedasticity) may not always hold.

Advantages of Multiple Linear Regression (MLR)

1. **Captures complex relationships** – Considers multiple predictors, giving better accuracy.
2. **Flexible** – Can model real-world problems where many factors influence the outcome.
3. **Identifies key variables** – Helps analyze which independent variables significantly affect the dependent variable.

Disadvantages of Multiple Linear Regression (MLR)

1. **Risk of multicollinearity** – High correlation among predictors can reduce reliability.
2. **Overfitting** – Too many predictors can make the model fit training data well but perform poorly on new data.
3. **Complex interpretation** – Coefficients are harder to explain compared to SLR.
4. **Data requirement** – Needs large, good-quality datasets to work effectively.

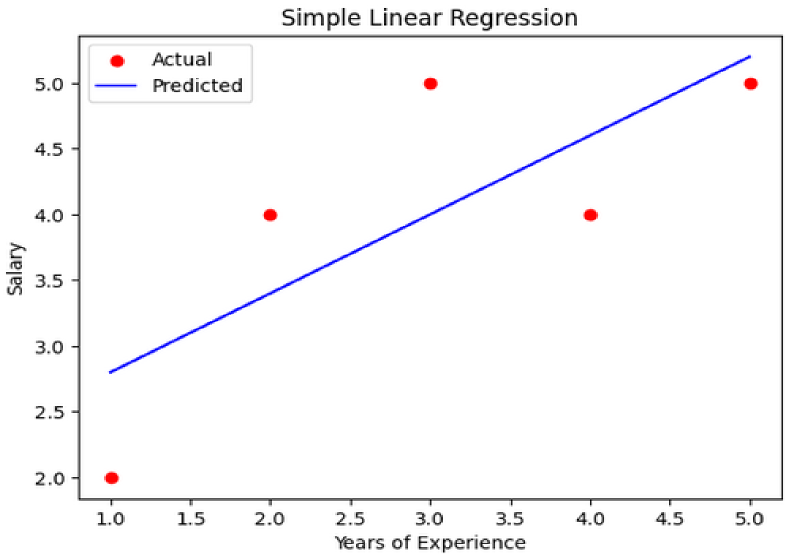
Program:

1) Simple Linear Regression:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# Sample dataset
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)   # Years of experience
y = np.array([2, 4, 5, 4, 5])                  # Salary (in LPA, example)

# Create model
model = LinearRegression()
model.fit(X, y)

# Predict
y_pred = model.predict(X)

# Results
print("Coefficient (slope):", model.coef_[0])
print("Intercept:", model.intercept_)

# Plot
plt.scatter(X, y, color='red', label="Actual")
plt.plot(X, y_pred, color='blue', label="Predicted")
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Simple Linear Regression")
plt.legend()
plt.show()
```

Output:

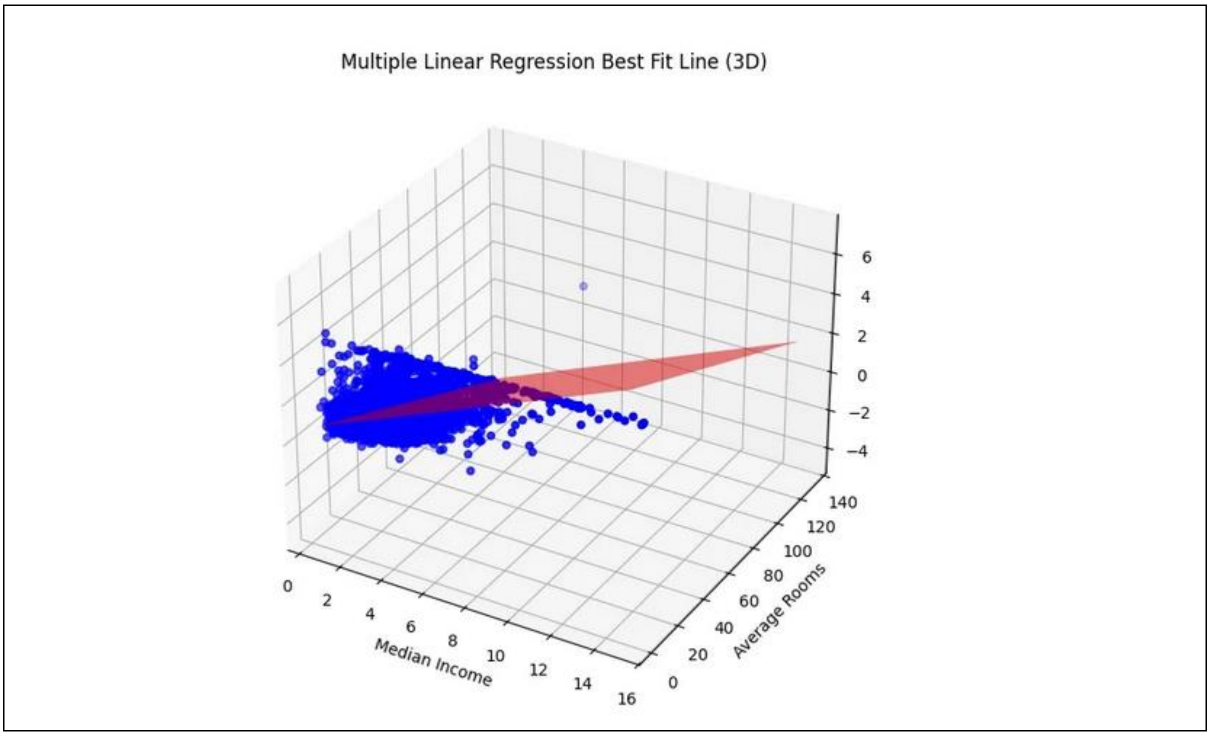
```
Coefficient (slope): 0.6
Intercept: 2.2
```



2) Multiple Linear Regression

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.datasets import fetch_california_housing
california_housing = fetch_california_housing()

X = pd.DataFrame(california_housing.data, columns=california_housing.feature_names)
y = pd.Series(california_housing.target)
X = X[['MedInc', 'AveRooms']]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')

ax.scatter(X_test['MedInc'], X_test['AveRooms'],
           y_test, color='blue', label='Actual Data')

x1_range = np.linspace(X_test['MedInc'].min(), X_test['MedInc'].max(), 100)
x2_range = np.linspace(X_test['AveRooms'].min(), X_test['AveRooms'].max(), 100)
x1, x2 = np.meshgrid(x1_range, x2_range)

z = model.predict(np.c_[x1.ravel(), x2.ravel()]).reshape(x1.shape)

ax.plot_surface(x1, x2, z, color='red', alpha=0.5, rstride=100, cstride=100)

ax.set_xlabel('Median Income')
ax.set_ylabel('Average Rooms')
ax.set_zlabel('House Price')
ax.set_title('Multiple Linear Regression Best Fit Line (3D)')

plt.show()
```

OUTPUT:



Conclusion:

Simple Linear Regression is used when the prediction depends on a single independent variable, while Multiple Linear Regression is used when multiple factors influence the prediction. Both help in understanding relationships between variables and making accurate forecasts.